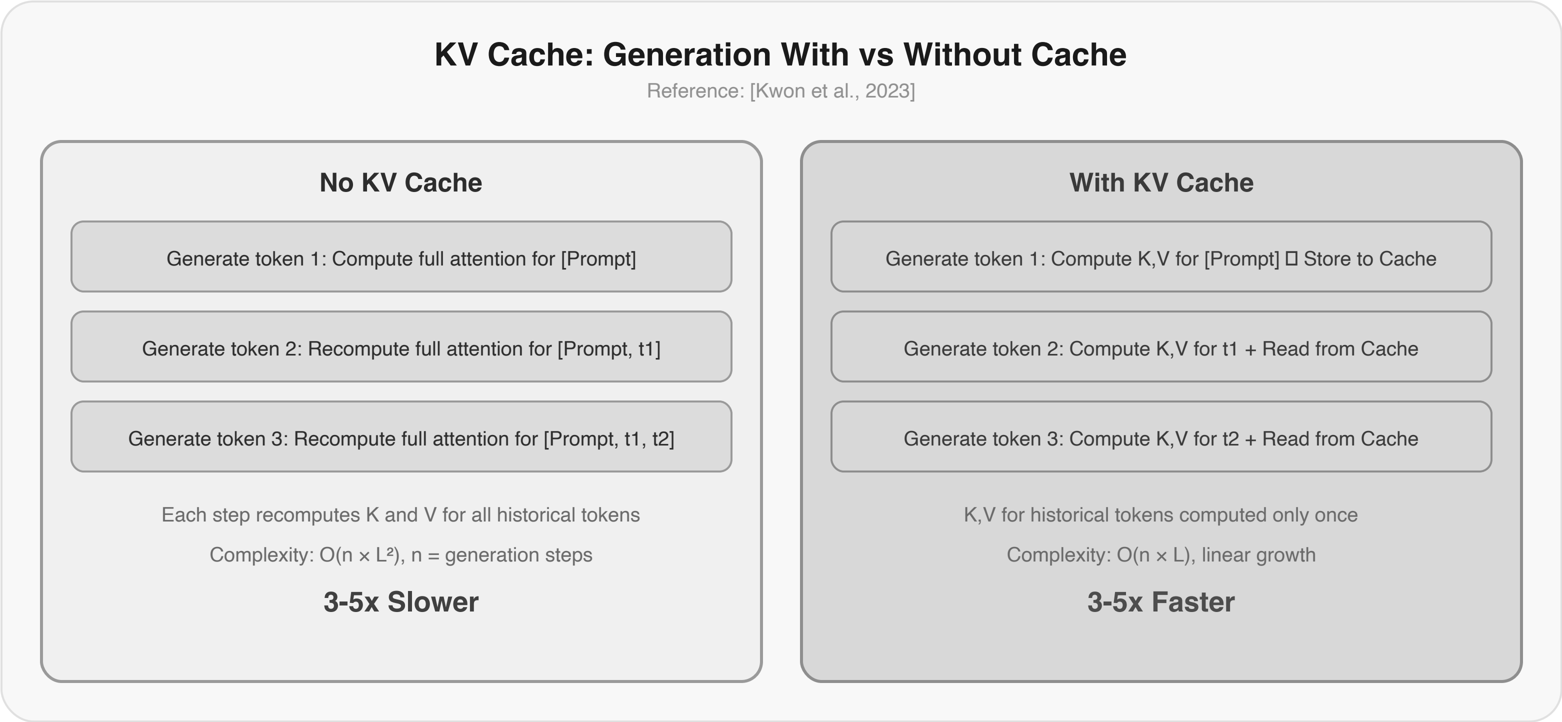
KV Cache: Generation With vs Without Cache
Reference: [Kwon et al., 2023]
No KV Cache
Generate token 1: Compute full attention for [Prompt]
Generate token 2: Recompute full attention for [Prompt, t1]
Generate token 3: Recompute full attention for [Prompt, t1, t2]
Each step recomputes K and V for all historical tokens
Complexity: O(n × L²), n = generation steps
3-5x Slower
With KV Cache
Generate token 1: Compute K,V for [Prompt] □ Store to Cache
Generate token 2: Compute K,V for t1 + Read from Cache
Generate token 3: Compute K,V for t2 + Read from Cache
K,V for historical tokens computed only once
Complexity: O(n × L), linear growth
3-5x Faster